

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 31

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well • Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well • Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate	3
01.2	• Natural gas may enter aquifer water • Injected chemicals may reach aquifers or the surface; toxic metals in the rock may be mobilised • Large volumes of water needed • Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) • Reduced by: collection, treatment and re-use of waste water; well casing standards; restricting sites away from aquifers / sensitive areas; monitoring	5
01.3	• Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices • Reserves only include what is economically exploitable now; high environmental impacts (land take, water, CO₂) also limit exploitation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	
02.1	• $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$	1
02.2	• Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ • Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter	2

Question	Marking guidance	Marks
02.3	<i>Any 2 from:</i> • Low / zero tillage • Adding manure, compost or mulch • Growing cover crops / leaving crop residues • Converting to long-term grass or tree crops in rotation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
03.1	• Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m^{-3}) • Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect)	2
03.2	• $(930 - 250) / 250 \times 100$ • = 272%	2
03.3	• Anticyclonic conditions: clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled • Low wind velocity so air layers did not mix; London lies in a valley / basin where cold dense air collected: a temperature inversion formed • Smoke from coal fires was trapped below the inversion; moist air cooled to its dew point forming fog; smoke + fog = smog; the smog's high albedo prolonged the inversion	3
03.4	• Smoke particles (PM10) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections • These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal	2
03.5	• Clean Air Act (1956)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
04.1	• $(150 - 125) / 150 \times 100 = 16.7\%$ (accept 17%)	1
04.2	• A fixed fee gives no incentive to save water because the bill is the same however much is used • Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances)	2
04.3	<i>Any 3 from:</i> • Low-water appliances: dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers • Greywater re-use: bath / shower water used to flush toilets or water gardens • Rainwater collection / water butts for garden use • Xeriscaping: planting drought-adapted species that need no watering • Repairing leaks / dripping taps	3
04.4	• Old infrastructure / pipes in poor condition leak • Repair and good maintenance / pipe replacement / leak detection reduce wastage	2
04.5	• Water is delivered directly to the plant roots / soil around each plant • Less evaporation loss from spray droplets and from wetting the whole soil surface; less runoff	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	

Question	Marking guidance	Marks
05.1	• A turbofan at the front forces bypass air around the inner cowling / core jet • The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise • The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest)	3
05.2	<i>Any 2 from:</i> • Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly • Acoustic liners in the engine cowlings absorb noise • Hush kits fitted to older low-bypass engines • Fairings on undercarriage and wing-flap hinges reduce turbulence; welded rather than riveted panels • Lighter composite aircraft need less thrust • Blended wing with engines on top so the body shields the ground	2
05.3	• Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer • Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low • Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft • More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters	4
05.4	• Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	
06.1	• $(44 - 8) / 50 = 0.72 \text{ m yr}^{-1}$	1
06.2	• Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation • The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines	2
06.3	• Porosity: a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) • Permeability: the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) • Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge	3
06.4	<i>Any 4 from:</i> • Salt water incursion: seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) • Subsidence: emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) • Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds • Ecological impacts: plants needing high water availability die; wetland animals lost; food chain effects • Reduced supplies: abstraction may suddenly have to stop after years of hidden decline	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
07.1	- Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen - It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury	2
07.2	- Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted - So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) - It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata)	3
07.3	- Mercury is persistent / not degraded; it is an element so cannot be broken down - It remains in the sediments and continues to be released / converted and taken up by the food chain; long-lived fish continue to accumulate it	2
07.4	<i>Any 3 from:</i> - Water current velocity and direction (dispersal) - Properties of the pollutant: solubility, persistence, liposolubility / potential to bioaccumulate - Food chain pathways to humans (fish, shellfish) / the critical group - Sediment type / adsorption onto clay - Volume of the water body / residence time / dilution - Sensitive locations downstream (fisheries, drinking water abstraction)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	
08.1	- Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports - Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones; open-cast mines take far more land than deep mines	2
08.2	- Dust: water sprays on roads / blasting areas make particles heavier and clump so they settle - Noise: baffle mounds / embankments absorb and deflect; blasting at set, predictable daytime hours; quieter machinery - Turbid water: sedimentation lagoons where water stands until solids settle before discharge - Leachate: collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals - Amenity: landscaping and tree planting to screen the mine; restoration plans agreed as a condition of planning permission	5
08.3	- Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface - At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village - Killing 144 people, mainly children in the primary school	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	

Question	Marking guidance	Marks
09.1	• Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive • Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected • Damage to other cell parts can stop the cell functioning or kill it; low damage may be repaired by the intact nucleus	3
09.2	• Acute: effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) • Chronic: effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts)	2
09.3	• Exposure: absorption of ionising radiation from a source the person is close enough to • Contamination: physically carrying radioactive material on or in the body, so exposure continues wherever the person goes; sealed sources can expose but not contaminate	2
09.4	• Sieverts = grays x radiation weighting factor (20 for alpha) • $0.002 \times 20 = 0.04 \text{ Sv}$ (40 mSv)	2
09.5	• The activity of a source (number of decays per second)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline • Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control) • Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields	9
10.2	<i>Any 1 from:</i> • Biological control (introduced predators / pathogens) • Predator habitats (hedgerows, beetle banks) • Crop rotation • Companion / barrier crops • Sterile male technique / pheromone traps • Genetic resistance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
11.1	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Stages of investigation: identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • Why Antarctica: low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • Effects of UVB: skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • Solutions: Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • Evaluation of success: ozone hole area / depth stabilising, CFC levels falling; limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier; role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	25